HIBBELER · ENGINEERING MECHANICS: STATICS, 14TH EDITION

STATICS IN TWELVE WEEKS

Study guide: the plan, reading assignments, notes, worked examples, formula cards and book problems for each week.

About 5 hours a week, weighted toward the statics in Transport Canada's Applied Mechanics exams (TP 2293): concurrent forces, beams, friction, belts and bearings, centre of gravity and hydrostatics.

The weekly tests and answer key are in the companion PDF. The interactive hub marks the tests for you and tracks your progress.

THE PLAN

Twelve weeks at a glance

WK	TOPIC	BOOK	ASSESSMENT	WEEK OF
1	Units, vectors and adding forces	Ch. 1 & 2.1–2.4	10-question test	_____
2	Equilibrium of a particle and free-body diagrams	Ch. 2.5–2.9 & 3	10-question test	_____
3	Moments of a force	Ch. 4.1–4.5	10-question test	_____
4	Couples, equivalent systems and distributed loads	Ch. 4.6–4.9	10-question test	_____
5	Equilibrium of a rigid body	Ch. 5	10-question test	_____
6	Trusses, frames and machines	Ch. 6	10-question test	_____
7	Internal forces: shear force and bending moment diagrams	Ch. 7	10-question test	_____
8	Dry friction and wedges	Ch. 8.1–8.3	10-question test	_____
9	Screws, belts, bearings and rolling resistance	Ch. 8.4–8.8	10-question test	_____
10	Centre of gravity, centroids and fluid pressure	Ch. 9	10-question test	_____
11	Moments of inertia and stability	Ch. 10 & 11	10-question test	_____
12	Review and final exam	All chapters	Final exam	_____

How each week runs (about 5 hours)

S1 · 90 min	**Read the book.** Core sections in full, skim the rest. Build a formula card as you go.
S2 · 90 min	**Notes and worked example.** Try the example on paper before reading the steps.
S3 · 60 min	**Book problems.** Preliminary, Fundamental and Review problems. Solutions are in the back: Fundamental p. 620, Preliminary p. 638, Review p. 648, selected answers p. 658.
S4 · 60 min	**Weekly test.** Closed book, calculator allowed. A week counts as passed at 70% or more.

Finding pages

All page numbers are the book's own printed numbers. In the PDF copy, **add 22** to get the viewer page: book p. 87 is PDF p. 109.

Where this fits

The Applied Mechanics exams list outcomes such as the resultant and equilibrant of coplanar forces, centre of gravity, friction on planes and in bearings, belt power, shear force and bending moment diagrams, and hydrostatic force on plane surfaces. Weeks 1–5 and 7–10 carry most of that. Weeks 6 and 11 build skills you'll lean on later. This book is statics only. The exams also cover dynamics, strength of materials, lifting machines and hydraulics, which are listed in week 12.

WEEK 1 · CH. 1 & 2.1–2.4

Units, vectors and adding forces

BY THE END OF THIS WEEK YOU CAN

- Work in SI units and prefixes, and convert mass to weight with W = mg
- Add two forces with the parallelogram law, using the cosine and sine rules
- Resolve any force into x and y components, including 3-4-5 slopes
- Find the resultant and the equilibrant of a set of coplanar forces

§	READING	PAGES	
1.1–1.6	General principles, units, numerical work	book pp. 3–16 (PDF pp. 25–38)	CORE
2.1–2.3	Scalars, vectors, parallelogram law	book pp. 17–32 (PDF pp. 39–54)	CORE
2.4	Adding a system of coplanar forces	book pp. 33–43 (PDF pp. 55–65)	CORE

	SESSION	WHAT TO DO	TIME
☐	**Read**	Core: §1.1–1.6, 2.1–2.3, 2.4. Copy each formula onto a cheat card.	90 min
☐	**Notes + worked example**	Work through the notes, then try the worked example before reading the steps.	90 min
☐	**Book problems**	P2–1, P2–2; F2–1 to F2–12; R2–1 to R2–4. Check every answer in the back of the book.	60 min
☐	**Weekly test**	10 questions, about 40 minutes, closed book. Pass mark 70%.	60 min

Notes

What statics is

Statics is the study of bodies that are at rest or moving at constant velocity, so the acceleration is zero. Every problem in this book comes down to one idea: **the forces on a body balance**. This week you learn to add forces correctly. Everything after this builds on that.

Newton's laws and weight

1st law: with no unbalanced force, a body stays at rest or keeps moving in a straight line. 2nd law: F = ma. 3rd law: action and reaction are equal and opposite. Weight is the force gravity puts on a mass: **W = mg**, with g = 9.81 m/s². A 1 kg mass weighs 9.81 N. A 1 tonne (1 Mg) mass weighs 9.81 kN.

SI units and prefixes

The newton is a derived unit: 1 N = 1 kg·m/s². Prefixes go in steps of 1000: kilo (k, 10^3), mega (M, 10^6), giga (G, 10^9), milli (m, 10^{-3}), micro (μ, 10^{-6}). Write 1 tonne as 1 Mg, never 1 kkg. Carry at least 4 figures through a calculation and round the final answer to 3 significant figures, as the book does.

Vectors

A **scalar** has size only (mass, time, length). A **vector** has size and direction (force, velocity, moment). You add two vectors with the **parallelogram law**: put them tail to tail, complete the parallelogram, and the diagonal is the resultant. The half-version is the **triangle rule**, which puts them head to tail. Solve the triangle with the cosine rule for the magnitude and the sine rule for the angle.

Resolving into components

For a force F at angle θ measured from the +x axis: $F_x = F \cos\theta$ and $F_y = F \sin\theta$. When the direction is given as a slope triangle, for example rise 3 and run 4 with hypotenuse 5, use the ratios: $F_x = F(4/5)$ and $F_y = F(3/5)$. Always check the sign of each component from a sketch. Don't trust the calculator to get the quadrant right.

Resultant of many forces

Add all the x-components and all the y-components: $F_{Rx} = \Sigma F_x$ and $F_{Ry} = \Sigma F_y$. Then $F_R = \sqrt{(F_{Rx}^2 + F_{Ry}^2)}$, and the direction is $\theta = \tan^{-1}(|F_{Ry}|/|F_{Rx}|)$, placed in the right quadrant by looking at the signs.

Equilibrant (exam term)

Hibbeler rarely uses the word, but Transport Canada exam papers do. The **equilibrant** is the single force that would hold the system in balance. It has the same magnitude as the resultant, acts along the same line, and points the opposite way, so its direction is $\theta_R + 180°$.

FORMULA CARD

Weight	$W = m\,g,\ g = 9.81\ m/s^2$
Cosine rule	$C = \sqrt{(A^2 + B^2 - 2AB\cos c)}$
Parallelogram (angle θ between the forces)	$R = \sqrt{(A^2 + B^2 + 2AB\cos\theta)}$
Sine rule	$A / \sin a = B / \sin b = C / \sin c$
Components	$F_x = F\cos\theta,\ F_y = F\sin\theta$
Resultant	$F_R = \sqrt{((\Sigma F_x)^2 + (\Sigma F_y)^2)},\ \theta = \tan^{-1}(\Sigma F_y / \Sigma F_x)$
Equilibrant	$E = F_R$, direction $\theta_R + 180°$

Worked example: Three lines on one bollard

Three mooring lines pull on one deck bollard. Take +x as forward and measure angles anticlockwise from +x: 5 kN at 0°, 8 kN at 60° and 4 kN at 150°. Find the resultant force on the bollard and the equilibrant that the bollard's foundation has to provide.

1. x-components: 5 cos 0° + 8 cos 60° + 4 cos 150° = 5 + 4 − 3.464 = **5.536 kN**
2. y-components: 5 sin 0° + 8 sin 60° + 4 sin 150° = 0 + 6.928 + 2 = **8.928 kN**
3. Magnitude: $F_R = \sqrt{(5.536^2 + 8.928^2)}$ = **10.5 kN**
4. Direction: both components are positive, so the resultant is in the 1st quadrant. $\theta = \tan^{-1}(8.928/5.536)$ = **58.2°**
5. Equilibrant: 10.5 kN acting at 58.2° + 180° = **238°**

Answer: F_R = 10.5 kN at 58.2°. Equilibrant = 10.5 kN at 238°.

ON THE SHIP

Two tugs on one tow, several mooring lines on a bollard, and a chain block hanging from a padeye that is pulled sideways are all coplanar force additions. The Applied Mechanics exams ask for the resultant and equilibrant of concurrent coplanar forces by name, so get fast and tidy at the component method this week.

Book problems

☐	**Preliminary**	`P2–1, P2–2`	book p. 27; solutions p. 638
☐	**Fundamental**	`F2–1 to F2–12`	book pp. 28 & 39; solutions p. 620
☐	**Review**	`R2–1 to R2–4`	book p. 84; solutions p. 648

WEEK 2 · CH. 2.5–2.9 & 3

Equilibrium of a particle and free-body diagrams

BY THE END OF THIS WEEK YOU CAN

- Draw a clean free-body diagram (FBD) of a particle
- Solve ΣFx = 0, ΣFy = 0 for unknown tensions
- Use Lami's theorem for three-force problems and see why wide sling angles are dangerous
- Handle springs (F = ks) and cables over pulleys
- Read a force written as a Cartesian vector (i, j, k), and find components along a line

§	READING	PAGES	
3.1–3.3	Equilibrium, free-body diagrams, coplanar systems	book pp. 87–105 (PDF pp. 109–127)	CORE
2.5–2.8	Cartesian vectors, position vectors, force along a line	book pp. 44–68 (PDF pp. 66–90)	SKIM
2.9	Dot product	book pp. 69–86 (PDF pp. 91–108)	SKIM
3.4	Three-dimensional force systems	book pp. 106–120 (PDF pp. 128–142)	SKIM

	SESSION	WHAT TO DO	TIME
☐	**Read**	Core: §3.1–3.3. Skim: §2.5–2.8, 2.9, 3.4. Copy each formula onto a cheat card.	90 min
☐	**Notes + worked example**	Work through the notes, then try the worked example before reading the steps.	90 min
☐	**Book problems**	P3–1, P3–2; F3–1 to F3–6 (core), F2–13 to F2–16 & F3–7, F3–8 (3D, optional); R3–1 to R3–4. Check every answer in the back of the book.	60 min
☐	**Weekly test**	10 questions, about 40 minutes, closed book. Pass mark 70%.	60 min

Notes

Equilibrium of a particle

A particle is in equilibrium when the resultant force on it is zero: **ΣF = 0**. In 2D that gives two equations, $\Sigma F_x = 0$ and $\Sigma F_y = 0$, so you can solve for at most **two unknowns**.

The free-body diagram

This is the most important skill in the course. (1) Draw the particle, such as a hook or a ring, isolated from everything around it. (2) Show *every* force acting on it: known loads, weights and the pulls of cables. (3) Label unknown magnitudes and mark every angle. If you assume a direction for an unknown and the answer comes out negative, the force acts the other way.

Cables, pulleys and springs

A cable can only **pull**, and its tension acts along the cable. Over a frictionless pulley the tension is the **same on both sides**. A linear spring has F = k·s, where s = deformed length − unstretched length. A positive s means the spring is stretched and pulling.

Lami's theorem (exam favourite)

When exactly three forces hold a point in equilibrium, each force is proportional to the sine of the angle between the other two: $\mathbf{F_1/\sin\alpha_1 = F_2/\sin\alpha_2 = F_3/\sin\alpha_3}$. It's the sine rule applied to the closed force triangle, and it is quick for hook-and-two-slings problems.

Sling angles

For a load W hung from two equal legs, each at angle θ from the vertical: T = W / (2 cos θ). At θ = 30° each leg carries 0.577W. At θ = 60° (a 120° included angle) each leg carries the **full** W. Wider is worse, which is why riggers limit the included angle.

Cartesian vectors (lighter touch)

In 3D, write $\mathbf{F} = F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k}$. To aim a force along a line from A to B: take the position vector $\mathbf{r} = (x_B - x_A)\mathbf{i} + \ldots$, divide it by its length to get the unit vector u, then F = F·u. Direction cosines satisfy $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$. The dot product A·B = AB cos θ gives the angle between two lines and the component of a force along a line. These exams are mostly 2D, so learn this well enough to follow it.

FORMULA CARD

Particle equilibrium	$\Sigma F_x = 0,\ \Sigma F_y = 0\ (\Sigma F_z = 0$ in 3D$)$
Lami's theorem	$F_1/\sin\alpha_1 = F_2/\sin\alpha_2 = F_3/\sin\alpha_3$
Symmetric two-leg sling	$T = W / (2\cos\theta)$ (θ from vertical)
Linear spring	$F = k\,s,\ s = L - L_0$
Force along a line	$\mathbf{F} = F\cdot(\mathbf{r} / \lvert\mathbf{r}\rvert)$
Direction cosines	$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$
Dot product	$\mathbf{A}\cdot\mathbf{B} = AB\cos\theta = A_xB_x + A_yB_y + A_zB_z$

Worked example: Lifting an engine block with two slings

A 2000 kg engine block hangs from a crane hook on two slings. Sling AB is 30° from the vertical, on the left. Sling AC is 45° from the vertical, on the right. Find the tension in each sling.

1. Weight: W = 2000 × 9.81 = **19.6 kN**.
2. FBD of the hook A: T_{AB} up and to the left at 30° from vertical, T_{AC} up and to the right at 45° from vertical, and W straight down.
3. $\Sigma F_x = 0$: $T_{AC}\sin 45° - T_{AB}\sin 30° = 0$
4. $\Sigma F_y = 0$: $T_{AB}\cos 30° + T_{AC}\cos 45° - W = 0$
5. Or use Lami directly. The angle opposite T_{AB} (between T_{AC} and W) is 180° − 45° = 135°. Opposite T_{AC} it is 180° − 30° = 150°. Opposite W it is 30° + 45° = 75°.
6. T_{AB} = W sin 135° / sin 75° = **14.4 kN**
7. T_{AC} = W sin 150° / sin 75° = **10.2 kN**

Answer: T_{AB} = 14.4 kN, T_{AC} = 10.2 kN. The sling nearer the vertical takes more of the load.

ON THE SHIP

This is chain-block and sling work in the engine room: lifting a cylinder head or a pump motor off padeyes. Every 'max load' problem here is really a safe-working-load check. The sling-angle rule, that each leg carries W at a 120° included angle, is worth knowing for the job as well as the exam.

Book problems

☐	**Preliminary**	P3–1, P3–2	book p. 96; solutions p. 638
☐	**Fundamental**	F3–1 to F3–6 (core), F2–13 to F2–16 & F3–7, F3–8 (3D, optional)	book pp. 51, 97, 111; solutions p. 620
☐	**Review**	R3–1 to R3–4	book p. 118; solutions p. 648

WEEK 3 · CH. 4.1–4.5

Moments of a force

BY THE END OF THIS WEEK YOU CAN

- Calculate a moment as force × perpendicular distance, with a sign
- Use Varignon's principle: the moment of a force equals the sum of the moments of its components
- Find a resultant moment from several forces
- Set up r × F for a simple 3D case

§	READING	PAGES	
4.1	Moment of a force, scalar form	book pp. 121–124 (PDF pp. 143–146)	CORE
4.4	Principle of moments (Varignon)	book pp. 132–144 (PDF pp. 154–166)	CORE
4.2–4.3	Cross product and vector moments	book pp. 125–131 (PDF pp. 147–153)	SKIM
4.5	Moment about a specified axis	book pp. 145–153 (PDF pp. 167–175)	SKIM

	SESSION	WHAT TO DO	TIME
☐	**Read**	Core: §4.1, 4.4. Skim: §4.2–4.3, 4.5. Copy each formula onto a cheat card.	90 min
☐	**Notes + worked example**	Work through the notes, then try the worked example before reading the steps.	90 min
☐	**Book problems**	P4–1 to P4–4; F4–1 to F4–12 (core), F4–13 to F4–18 (optional); R4–1, R4–3. Check every answer in the back of the book.	60 min
☐	**Weekly test**	10 questions, about 40 minutes, closed book. Pass mark 70%.	60 min

Notes

What a moment is

A moment is the turning effect of a force about a point or an axis. In scalar form, $\mathbf{M_O = F \cdot d}$, where d is the **perpendicular** distance from O to the force's *line of action*. The units are N·m or kN·m. Don't confuse a moment with work (J). They share base units, but they are different things.

Sign convention

Pick one convention and keep to it. The book uses **anticlockwise (CCW) = positive**, which is the right-hand rule with the thumb out of the page. A force whose line of action passes through O has zero moment about O.

Principle of moments (Varignon)

The moment of a force about a point equals the sum of the moments of its components about that point. In practice, split F into F_x and F_y, which are easy because their perpendicular distances are just coordinates: $\mathbf{M_O = x \cdot F_y - y \cdot F_x}$. This is often faster than finding d.

Resultant moment

Add the moments of all the forces about the same point with their signs: $(M_R)_O = \Sigma F \cdot d$. A negative answer means the net turning is clockwise.

Vector form (lighter touch)

$\mathbf{M_O = r \times F}$, where r runs from O to *any* point on the line of action. Work out the 3×3 determinant with rows i j k / r_x r_y r_z / F_x F_y F_z. The moment about an axis with unit vector u is $M_a = u \cdot (r \times F)$. It matters for shafts and hinges, but exam questions are mostly planar.

FORMULA CARD

Scalar moment	$M_O = F\ d$ ($d \perp$ to line of action)
Force at an angle φ to a lever of length L	$M = F\ L \sin \varphi$
Varignon (2D)	$M_O = x\ F_y - y\ F_x$
Resultant moment	$(M_R)_O = \Sigma\ F\ d$ (CCW +)
Vector moment	$M_O = r \times F$
Moment about an axis	$M_a = u_a \cdot (r \times F)$

Worked example: Torque on a flange bolt, two ways

(a) An engineer pulls on a 0.3 m spanner with 400 N at 75° to the handle. What torque does the bolt get? (b) A 2 kN force with components F_x = 1.2 kN and F_y = 1.6 kN acts at the point (3, 2) m. Find its moment about the origin O.

1. (a) Only the component perpendicular to the handle turns the bolt: M = F L sin φ = 400 × 0.3 × sin 75° = **116 N·m**.
2. (b) Use Varignon with components. F_y = 1.6 kN acts at x = 3 m and turns anticlockwise (+). F_x = 1.2 kN acts at y = 2 m and turns clockwise (–).
3. $M_O = x\ F_y - y\ F_x = 3(1.6) - 2(1.2) = 4.8 - 2.4 =$ **2.40 kN·m**, anticlockwise.

Answer: (a) 116 N·m. (b) 2.40 kN·m anticlockwise.

ON THE SHIP

This covers torquing flange and cylinder-head bolts, valve hand wheels, the turning moment on a rudder stock, and the overturning moment a provisions crane puts on its pedestal. 'Force × perpendicular distance' is behind every lever, crank and tiller you'll meet.

Book problems

☐	**Preliminary**	P4–1 to P4–4	book pp. 135 & 150; solutions p. 638
☐	**Fundamental**	F4–1 to F4–12 (core), F4–13 to F4–18 (optional)	book pp. 136–137, 151; solutions p. 620
☐	**Review**	R4–1, R4–3	book p. 204; solutions p. 648

WEEK 4 · CH. 4.6–4.9

Couples, equivalent systems and distributed loads

BY THE END OF THIS WEEK YOU CAN

- Recognise a couple and calculate its moment
- Move a force to another point by adding a couple
- Replace parallel forces with a single resultant and find where it acts
- Turn uniform, triangular and trapezoidal loads into equivalent point loads

§	READING	PAGES	
4.6	Moment of a couple	book pp. 154–165 (PDF pp. 176–187)	CORE
4.7–4.8	Simplifying force and couple systems	book pp. 166–189 (PDF pp. 188–211)	CORE
4.9	Reducing a simple distributed load	book pp. 190–206 (PDF pp. 212–228)	CORE

	SESSION	WHAT TO DO	TIME
☐	**Read**	Core: §4.6, 4.7–4.8, 4.9. Copy each formula onto a cheat card.	90 min
☐	**Notes + worked example**	Work through the notes, then try the worked example before reading the steps.	90 min
☐	**Book problems**	P4–5 to P4–7; F4–19 to F4–21, F4–25 to F4–28, F4–31 to F4–34, F4–37 to F4–40; R4–5 to R4–8. Check every answer in the back of the book.	60 min
☐	**Weekly test**	10 questions, about 40 minutes, closed book. Pass mark 70%.	60 min

Notes

Couples

A **couple** is two parallel forces that are equal in size, opposite in direction, and a distance d apart. They have zero resultant force but a turning effect of **M = F·d**. A couple moment is a **free vector**: it has the same value about every point, so you can move it anywhere on the body.

Moving a force

You can slide a force along its own line of action without changing anything. This is transmissibility. To move it *off* its line to a point O, add a couple equal to the moment the force had about O: M = F·d. A force-and-couple system at O is equivalent to the original force.

Resultant of parallel forces

For vertical loads on a beam: $F_R = \Sigma F$. Find its position by matching moments: **$\bar{x} = \Sigma(F \cdot x) / \Sigma F$**. This is the same idea as finding a centre of gravity, which comes in week 10.

Distributed loads

A load spread along a length, w in kN/m, is equivalent to a single force equal to the **area under the load diagram**, acting through the **centroid** of that area. Uniform (UDL): F = wL at L/2. Triangular from 0 to w_0: F = ½w_0L at 2L/3 from the zero end, which is L/3 from the heavy end. Treat a trapezoid as a rectangle plus a triangle.

FORMULA CARD

Couple moment	$M = F\ d$ (free vector)
Force moved to O	F at O + couple $M = F\ d$
Parallel resultant	$F_R = \Sigma F$, $\bar{x} = \Sigma(F\ x) / \Sigma F$
Uniform load	$F = wL$ at $L/2$
Triangular load	$F = \frac{1}{2}\ w_0\ L$ at $L/3$ from the w_0 end
Any distributed load	$F_R = \int w\ dx$, $\bar{x} = \int x\ w\ dx / \int w\ dx$

Worked example: Deck beam under stores and a pallet

A 6 m deck beam AB carries a uniform load of stores of 4 kN/m along its full length, plus a 10 kN pallet 2 m from A. Replace all of this with a single resultant force and give its position from A.

1. Uniform load → 4 × 6 = 24 kN acting at 3 m from A.
2. Total: F_R = 24 + 10 = **34 kN**, downward.
3. Moments about A: 24(3) + 10(2) = 72 + 20 = 92 kN·m.
4. $\bar{x}$ = 92 / 34 = **2.71 m** from A.

Answer: F_R = 34 kN downward, acting 2.71 m from A.

ON THE SHIP

Turning distributed loads into point loads is the first step in every beam problem: cargo on a hatch cover, tank contents on a floor plate, or water pressure on a bulkhead, which is a triangular load. Propeller torque and a steering-gear rudder moment are couples. Next week you use these resultants to find support reactions.

Book problems

☐	**Preliminary**	P4–5 to P4–7	book pp. 172 & 184; solutions p. 638
☐	**Fundamental**	F4–19 to F4–21, F4–25 to F4–28, F4–31 to F4–34, F4–37 to F4–40	book pp. 160, 173, 185, 195; solutions p. 620
☐	**Review**	R4–5 to R4–8	book p. 204; solutions p. 648

WEEK 5 · CH. 5

Equilibrium of a rigid body

BY THE END OF THIS WEEK YOU CAN

- Draw the correct reactions for rollers, pins and fixed supports
- Solve the three equilibrium equations ΣFx = 0, ΣFy = 0, ΣM = 0
- Choose moment points that remove unknowns
- Recognise two-force members and use them to simplify problems
- Tell whether a structure is statically determinate

§	READING	PAGES	
5.1–5.2	Conditions for equilibrium, free-body diagrams	book pp. 207–219 (PDF pp. 229–241)	CORE
5.3–5.4	Equations of equilibrium, two- and three-force members	book pp. 220–244 (PDF pp. 242–266)	CORE
5.5–5.7	3D equilibrium, constraints, statical determinacy	book pp. 245–272 (PDF pp. 267–294)	SKIM

	SESSION	WHAT TO DO	TIME
☐	**Read**	Core: §5.1–5.2, 5.3–5.4. Skim: §5.5–5.7. Copy each formula onto a cheat card.	90 min
☐	**Notes + worked example**	Work through the notes, then try the worked example before reading the steps.	90 min
☐	**Book problems**	P5–1; F5–1 to F5–6; R5–1 to R5–4. Check every answer in the back of the book.	60 min
☐	**Weekly test**	10 questions, about 40 minutes, closed book. Pass mark 70%.	60 min

Notes

Three equations

A rigid body in 2D is in equilibrium when $\mathbf{\Sigma F_x = 0}$, $\mathbf{\Sigma F_y = 0}$ **and** $\mathbf{\Sigma M_O = 0}$. That gives three equations, so at most three unknowns. Take moments about a point where two unknowns act, so that each equation has only one unknown.

Support reactions

A support pushes back in each direction it stops the body moving. **Roller or smooth surface**: 1 force, perpendicular to the surface. **Pin or hinge**: 2 force components, A_x and A_y. **Fixed support** (built-in, welded): 2 forces plus a moment, M_A. A cable gives 1 tension along the cable.

Free-body diagram for a rigid body

Draw the body on its own. Replace every support with its reactions. Add the weight at the centre of gravity and replace each distributed load with its resultant (week 4). Show all the dimensions you'll need for moments. An unknown that comes out negative just acts the other way.

Two- and three-force members

A **two-force member** has forces applied at only two points and no weight. For equilibrium those two forces must be equal, opposite and **along the line joining the two points**. Links, struts and topping lifts are like this, so their direction is known straight away. In a **three-force member**, the three forces must meet at one point or all be parallel.

Determinacy and constraints

When there are more unknown reactions than equations, the body is **statically indeterminate**, like a shaft on three bearings. Statics alone can't solve it. When reactions are all parallel or all meet at one point, the body is **improperly constrained** and can move even if there are enough supports.

FORMULA CARD

2D equilibrium	$\Sigma F_x = 0,\ \Sigma F_y = 0,\ \Sigma M_O = 0$
Roller / pin / fixed	1 / 2 / 3 unknowns (fixed adds M)
Simply supported, point load P at a from A (span L)	$R_B = P\,a / L,\ R_A = P\,(L - a)/L$
Cantilever, end load P	$R = P,\ M_{wall} = P\,L$
Cantilever, UDL w	$R = wL,\ M_{wall} = wL^2/2$
Lever	$F_1\,d_1 = F_2\,d_2$

Worked example: Reactions on a pump skid beam

A simply supported beam AB spans 6 m, with a pin at A and a roller at B. It carries a 12 kN point load 2 m from A and a uniform load of 3 kN/m over the last 2 m, from x = 4 m to 6 m. Find the reactions.

1. Resultant of the uniform load: 3 × 2 = 6 kN, acting at x = 5 m.
2. There are no horizontal loads, so $A_x = 0$.
3. $\Sigma M_A = 0$: $R_B(6) - 12(2) - 6(5) = 0$, so $R_B = 54/6 =$ **9.00 kN**.
4. $\Sigma F_y = 0$: $R_A + R_B - 12 - 6 = 0$, so $R_A =$ **9.00 kN**.
5. Check with $\Sigma M_B = 0$: $-9(6) + 12(4) + 6(1) = -54 + 48 + 6 = 0$ ✓

Answer: R_A = 9.00 kN up, $A_x = 0$, R_B = 9.00 kN up.

ON THE SHIP

This covers bearing loads on a line shaft, reactions under a pump or motor sitting on a skid, the pin forces at a derrick heel, and the tension in a topping lift. The topping lift is a classic two-force member. Next week's trusses and the week 7 bending moments both start from reactions, so get fluent here.

Book problems

☐	**Preliminary**	P5–1	book p. 232; solutions p. 638
☐	**Fundamental**	F5–1 to F5–6	book p. 233; solutions p. 620
☐	**Review**	R5–1 to R5–4	book p. 270; solutions p. 648

WEEK 6 · CH. 6

Trusses, frames and machines

BY THE END OF THIS WEEK YOU CAN

- State the assumptions behind truss analysis
- Find member forces by the method of joints and label them tension or compression
- Spot zero-force members by inspection
- Cut a section and use one moment equation to get a member force directly
- Take apart a frame or machine and find pin forces and mechanical advantage

§	READING	PAGES	
6.1–6.3	Simple trusses, method of joints, zero-force members	book pp. 273–290 (PDF pp. 295–312)	CORE
6.4	Method of sections	book pp. 291–300 (PDF pp. 313–322)	CORE
6.6	Frames and machines	book pp. 305–342 (PDF pp. 327–364)	CORE
6.5	Space trusses	book pp. 301–304 (PDF pp. 323–326)	SKIP

	SESSION	WHAT TO DO	TIME
☐	**Read**	Core: §6.1–6.3, 6.4, 6.6. Copy each formula onto a cheat card.	90 min
☐	**Notes + worked example**	Work through the notes, then try the worked example before reading the steps.	90 min
☐	**Book problems**	P6–1 to P6–3; F6–1 to F6–12, F6–17 to F6–20; R6–1, R6–2, R6–3, R6–6, R6–8. Check every answer in the back of the book.	60 min
☐	**Weekly test**	10 questions, about 40 minutes, closed book. Pass mark 70%.	60 min

Notes

Truss assumptions

Members are joined by smooth pins, loads act only at the joints, and member weights are ignored or shared out to the joints. That makes every member a **two-force member** carrying pure **tension (T)** or **compression (C)** along its length.

Method of joints

Find the support reactions first. Then go joint by joint, starting at one with **no more than two unknowns**, using $\Sigma F_x = 0$ and $\Sigma F_y = 0$. Assume every unknown member is in tension, pulling away from the joint. A negative answer means compression.

Zero-force members

(1) At a joint with only two members that aren't in line and no load, **both** members carry zero. (2) At a joint with three members where two are in line and there's no load, the **third** member carries zero. Spot these first to save work.

Method of sections

Cut the truss through the member you want, cutting **no more than three** unknown members. Take one side as a free body. Take moments about the point where the other two cut members meet, and the force you want falls out of one equation.

Frames and machines

These contain at least one **multi-force member**, so the members are no longer simple two-force members. Take the structure apart. At each pin, the forces on the two members are equal and opposite (Newton's 3rd law). Spot any two-force members first, since their direction is known. A machine changes an input force into an output force. Its **mechanical advantage** is output ÷ input.

FORMULA CARD

Each joint	$\Sigma F_x = 0,\ \Sigma F_y = 0$
Each section	$\Sigma F_x = 0,\ \Sigma F_y = 0,\ \Sigma M = 0$ ($\leq$ 3 cut members)
Member force from its y-part	$F = F_y \times (\text{length} / \text{rise})$
Chord force by sections	$F_{chord} = M_{section} / \text{depth}$
Mechanical advantage	$MA = F_{out} / F_{in}$

Worked example: A triangular jib frame

A simple triangular truss has supports A (pin) and C (roller) 6 m apart. Its apex B is 4 m above the mid-point of AC, and a 10 kN load hangs from B. Find the forces in AB and AC.

1. Reactions: the truss is symmetric, so $A_y = C_y = 5$ kN and $A_x = 0$.
2. Geometry of AB: run 3 m, rise 4 m, length 5 m, at $\theta = 53.1°$ to the horizontal.
3. Joint A, ΣF_y: $5 + F_{AB} \sin\theta = 0$, so $F_{AB} = -5 \times 5/4 =$ **-6.25 kN**. The minus sign means it is **compression**.
4. Joint A, ΣF_x: $F_{AC} + F_{AB} \cos\theta = 0$, so $F_{AC} = 6.25 \times 3/5 =$ **3.75 kN (T)**.

Answer: $F_{AB} = 6.25$ kN (C), $F_{AC} = 3.75$ kN (T). By symmetry BC = 6.25 kN (C).

ON THE SHIP

Cargo derricks, crane jibs, A-frames, overhead lifting beams with their hangers, and hatch-cover frames are all trusses or frames. Most engine-room tools are machines: chain blocks, pipe wrenches, valve-spindle levers. Trusses get less weight than beams and friction in the Applied Mechanics exams, so treat this as solid practice in equilibrium rather than something to memorise.

Book problems

☐	**Preliminary**	P6–1 to P6–3	book pp. 285 & 321; solutions p. 638
☐	**Fundamental**	F6–1 to F6–12, F6–17 to F6–20	book pp. 286, 297, 322–324; solutions p. 620
☐	**Review**	R6–1, R6–2, R6–3, R6–6, R6–8	book p. 340; solutions p. 648

WEEK 7 · CH. 7

Internal forces: shear force and bending moment diagrams

BY THE END OF THIS WEEK YOU CAN

- Find the internal normal force N, shear V and moment M at a section
- Write V(x) and M(x) for a beam and draw the diagrams
- Use dV/dx = w and dM/dx = V to sketch diagrams quickly
- Find the maximum bending moment and where it occurs
- Use the standard results for cantilevers and simply supported beams

§	READING	PAGES	
7.1	Internal loadings in members	book pp. 343–360 (PDF pp. 365–382)	CORE
7.2	Shear and moment equations and diagrams	book pp. 361–369 (PDF pp. 383–391)	CORE
7.3	Relations between load, shear and moment	book pp. 370–380 (PDF pp. 392–402)	CORE
7.4	Cables	book pp. 381–400 (PDF pp. 403–422)	SKIM

	SESSION	WHAT TO DO	TIME
☐	**Read**	Core: §7.1, 7.2, 7.3. Skim: §7.4. Copy each formula onto a cheat card.	90 min
☐	**Notes + worked example**	Work through the notes, then try the worked example before reading the steps.	90 min
☐	**Book problems**	P7–1; F7–1 to F7–4, F7–7 to F7–12, F7–13 to F7–16; R7–1 to R7–5. Check every answer in the back of the book.	60 min
☐	**Weekly test**	10 questions, about 40 minutes, closed book. Pass mark 70%.	60 min

Notes

Why this matters

Members fail where the internal bending moment is largest. Shear force and bending moment (SF and BM) diagrams for cantilevers and simply supported beams with point loads and uniform loads are a **named outcome** in the Applied Mechanics exams. Spend extra time here.

Internal loadings

Cut the member at the section you're interested in. Either piece is in equilibrium, so the cut face must carry a normal force **N**, a shear force **V** and a bending moment **M**. Find them with the three equilibrium equations on one piece, usually the simpler one.

Sign convention (Hibbeler)

Positive V makes the segment rotate clockwise. Positive M makes the beam **sag** (concave up), with the top in compression. A cantilever under downward load has **negative** moment, which is hogging. Exam papers sometimes use other conventions, so state yours and keep to it.

The quick-sketch rules

Slope of the shear diagram = load intensity: dV/dx = w, with w upward positive. So a UDL gives a sloping straight line, and a point load makes the shear diagram jump by the size of the load. **Slope of the moment diagram = shear**: dM/dx = V. So ΔM between two points = the area under the shear diagram. **M is a maximum where V = 0** or changes sign.

Standard cases to memorise

Simply supported, central point load P: $M_{max} = PL/4$ at mid-span. Simply supported, UDL w: $M_{max} = wL^2/8$ at mid-span. Cantilever, end load P: $M_{max} = PL$ at the wall. Cantilever, UDL w: $M_{max} = wL^2/2$ at the wall.

Hogging and sagging of a ship

A ship's hull is a beam. Weight and buoyancy are distributed differently along the length, and that difference is the load. With heavy weight amidships, or a wave trough amidships, the hull **sags**: the deck is in compression and the keel in tension. With weight at the ends, or a wave crest amidships, it **hogs**. This is the same SF and BM idea on the biggest beam you'll ever work on.

FORMULA CARD

Load–shear	$dV/dx = w$ (jump of P at a point load)
Shear–moment	$dM/dx = V$, ΔM = area under V diagram
SS beam, central P	$M_{max} = PL/4$
SS beam, UDL	$M_{max} = wL^2/8$, $V_{max} = wL/2$
SS beam, P at a from A	$M_{max} = P\,a\,(L - a) / L$
Cantilever, end P	$M_{max} = PL$, $V = P$
Cantilever, UDL	$M_{max} = wL^2/2$, $V_{max} = wL$

Worked example: A lifting beam under a uniform load

A simply supported beam spans 8 m and carries a uniform load of 5 kN/m. Write the equations for V(x) and M(x), and find the maximum moment.

1. Reactions: $R_A = R_B = 5 \times 8/2 = 20$ kN.
2. Cut at x from A. From the left-hand piece: $V(x) = 20 - 5x$ kN.
3. $M(x) = 20x - 5x\cdot(x/2) = 20x - 2.5x^2$ kN·m.
4. $V = 0$ where $20 - 5x = 0$, so $x = 4$ m, which is mid-span.
5. $M_{max} = 20(4) - 2.5(16) =$ **40 kN·m**. Check: $wL^2/8 = 5 \times 64/8 = 40$ ✓
6. Shape: V is a straight line from +20 to −20 kN. M is a parabola, 0 at each end and 40 at mid-span.

Answer: $V(x) = 20 - 5x$ kN, $M(x) = 20x - 2.5x^2$ kN·m, $M_{max} = 40$ kN·m at $x = 4$ m.

ON THE SHIP

Line shafting between bearings, lifting beams, deck girders and the hull girder itself are all beams. SF and BM diagrams for cantilevers and simply supported beams are a listed Applied Mechanics exam outcome, and the same diagrams sit behind hogging and sagging in ship stability.

Book problems

☐	**Preliminary**	P7–1	book p. 351; solutions p. 638
☐	**Fundamental**	F7–1 to F7–4, F7–7 to F7–12, F7–13 to F7–16	book pp. 352, 365, 376; solutions p. 620
☐	**Review**	R7–1 to R7–5	book p. 399; solutions p. 648

WEEK 8 · CH. 8.1–8.3

Dry friction and wedges

BY THE END OF THIS WEEK YOU CAN

- Use $F \le \mu sN$, and know when friction is at its limit
- Relate the angle of friction and angle of repose to μ
- Solve block-on-incline problems, pushing up and holding back
- Decide whether a body slips or tips first
- Calculate the work done against friction moving up a plane

§	READING	PAGES	
8.1	Characteristics of dry friction	book pp. 401–405 (PDF pp. 423–427)	CORE
8.2	Problems involving dry friction	book pp. 406–429 (PDF pp. 428–451)	CORE
8.3	Wedges	book pp. 430–431 (PDF pp. 452–453)	CORE

	SESSION	WHAT TO DO	TIME
☐	**Read**	Core: §8.1, 8.2, 8.3. Copy each formula onto a cheat card.	90 min
☐	**Notes + worked example**	Work through the notes, then try the worked example before reading the steps.	90 min
☐	**Book problems**	P8–1 to P8–4; F8–1 to F8–9; R8–1, R8–2, R8–3, R8–5. Check every answer in the back of the book.	60 min
☐	**Weekly test**	10 questions, about 40 minutes, closed book. Pass mark 70%.	60 min

Notes

The friction law

Friction acts along the contact surface and opposes motion, or the motion that is about to happen. Static friction adjusts to whatever is needed, up to a limit: $\mathbf{F \le \mu_s N}$. At the point of slipping, $F = \mu_s N$. Once sliding, $\mathbf{F_k = \mu_k N}$, and μ_k is usually about 25% less than μ_s. Friction does not depend on contact area.

Angle of friction

The resultant of N and F leans at angle φ from the normal, where $\mathbf{\tan \varphi_s = \mu_s}$. A block on an incline starts to slide when the slope reaches φ_s. This is the **angle of repose**.

Inclined plane (exam staple)

Weight W on a slope θ, with the force P parallel to the slope. **Push up** at impending motion or steady speed: $P = W(\sin\theta + \mu\cos\theta)$. **Hold from sliding down**: $P = W(\sin\theta - \mu\cos\theta)$. If this comes out negative, the block holds itself. Work done moving distance s up the slope is $P{\cdot}s$, and the friction part of it is $\mu W\cos\theta{\cdot}s$.

Slip or tip?

A tall box pushed sideways can slide or tip over. Work out the push needed for each. Slip: $P = \mu_s W$. Tip about the front edge: $P{\cdot}h = W{\cdot}(b/2)$. The **smaller** P is what happens first. Then check that N and F are possible: $N \ge 0$, and $F \le \mu N$ unless the box is slipping.

Wedges

Wedges lift or adjust heavy loads with a small force. Draw an FBD of each piece, with friction on **every** sliding face opposing the motion that is about to happen. A wedge is **self-locking**, so it stays in place when P is removed, if the friction is high enough compared with the wedge angle.

FORMULA CARD

Static (limit) / kinetic	$F_s = \mu_s N$, $F_k = \mu_k N$
Angle of friction	$\tan \varphi = \mu$
Push up a slope (P ∥ slope)	$P = W(\sin\theta + \mu\cos\theta)$
Hold on a slope (P ∥ slope)	$P = W(\sin\theta - \mu\cos\theta)$
Work up a slope	$U = P\,s$
Tipping (push at height h, base b)	$P_{tip} = W\,b / (2h)$

Worked example: Skidding a crate up a ramp

A 200 kg crate of spares sits on a 20° ramp with $\mu_s = 0.30$ and $\mu_k = 0.25$. (a) Will it stay put on its own? (b) What force, parallel to the ramp, keeps it moving up at steady speed?

1. W = 200 × 9.81 = 1 962 N.
2. (a) Compare the slope with the angle of friction: tan 20° = 0.364, which is more than $\mu_s = 0.30$. The crate **slides down** unless it's held.
3. (b) Moving up, so kinetic friction acts down the slope: $P = W(\sin 20° + \mu_k \cos 20°)$.
4. P = 1 962(0.3420 + 0.25 × 0.9397) = **1 132 N**.

Answer: (a) It slides, because tan 20° > μ_s. (b) P = 1 132 N.

ON THE SHIP

Chocks and wedges under machinery, skidding a pump across deck plates, cargo shifting as the ship rolls, and hand-ramping a drum up a gangway are all friction problems. The Applied Mechanics outcome 'work done moving up a plane at uniform speed' is exactly the inclined-plane formula from this week.

Book problems

☐	**Preliminary**	P8–1 to P8–4	book p. 415; solutions p. 638
☐	**Fundamental**	F8–1 to F8–9	book pp. 416–417; solutions p. 620
☐	**Review**	R8–1, R8–2, R8–3, R8–5	book p. 462; solutions p. 648

WEEK 9 · CH. 8.4–8.8

Screws, belts, bearings and rolling resistance

BY THE END OF THIS WEEK YOU CAN

- Find the torque to raise or lower a load with a square-threaded screw, and check for self-locking
- Use the belt and capstan equation $T_2 = T_1 e^{\mu\beta}$
- Calculate the power a belt drive transmits
- Find the friction torque and power lost in collar and journal bearings
- Estimate rolling resistance

§	READING	PAGES	
8.4	Frictional forces on screws	book pp. 432–438 (PDF pp. 454–460)	CORE
8.5	Frictional forces on flat belts	book pp. 439–446 (PDF pp. 461–468)	CORE
8.6	Collar bearings, pivot bearings and disks	book pp. 447–449 (PDF pp. 469–471)	CORE
8.7	Journal bearings	book pp. 450–451 (PDF pp. 472–473)	CORE
8.8	Rolling resistance	book pp. 452–461 (PDF pp. 474–483)	SKIM

	SESSION	WHAT TO DO	TIME
☐	**Read**	Core: §8.4, 8.5, 8.6, 8.7. Skim: §8.8. Copy each formula onto a cheat card.	90 min
☐	**Notes + worked example**	Work through the notes, then try the worked example before reading the steps.	90 min
☐	**Book problems**	8–71, 8–72, 8–77; 8–83, 8–84, 8–89, 8–91; 8–107, 8–108, 8–116, 8–117, 8–120; R8–4, R8–6, R8–8. Check every answer in the back of the book.	60 min
☐	**Weekly test**	10 questions, about 40 minutes, closed book. Pass mark 70%.	60 min

Notes

Square-threaded screws

A screw thread is an incline wrapped around a cylinder. The lead angle is $\theta = \tan^{-1}(l / 2\pi r)$, with l the lead and r the mean radius. The torque to **raise** load W is $M = W r \tan(\theta + \varphi_s)$, where $\varphi_s = \tan^{-1} \mu_s$. The screw is **self-locking**, so it holds the load with no torque applied, when $\varphi_s > \theta$. The torque to lower a self-locking screw is $M = W r \tan(\varphi_s - \theta)$.

Belts and ropes on drums

A flat belt or rope wrapped through an angle β (in **radians**) on a drum, just about to slip: $\mathbf{T_2 = T_1 e^{\mu\beta}}$. T_2 is the larger tension, on the side being pulled toward the motion. One full turn is 2π rad. Because the relation is exponential, a few turns let one deckhand hold a ship.

Belt power

A belt drive transmits power through the *difference* in tension: $\mathbf{P = (T_2 - T_1)\cdot v}$, with v the belt speed in m/s. For V-belts the effective μ is larger, because the belt wedges into the groove. This is covered in the exam syllabus, but Hibbeler treats only flat belts.

Collar and pivot bearings

A thrust collar or pivot carries an **axial** load P. Assuming uniform pressure, the friction torque is $M = (2/3)\,\mu\,P\,(R_2^3 - R_1^3)/(R_2^2 - R_1^2)$. For a solid disk ($R_1 = 0$), $M = (2/3)\,\mu\,P\,R$.

Journal bearings and power lost

A shaft in a journal bearing carries a **radial** load P. The friction torque is M = P r sin $\varphi_k \approx$ **μ P r** when μ is small. The power lost to friction is **M·ω**, with ω = 2πN/60 rad/s. This is a named exam outcome: 'energy and power lost to friction in simple bearings'.

Rolling resistance

A wheel on a soft surface needs a push P ≈ W a / r to keep it rolling, where a is the coefficient of rolling resistance (a length) and r is the wheel radius.

FORMULA CARD

Screw lead angle	$\theta = \tan^{-1}(l / 2\pi r)$
Screw raise / lower	$M = W r \tan(\theta + \varphi)$, $M = W r \tan(\varphi - \theta)$
Self-locking	$\varphi_s > \theta$
Belt / capstan	$T_2 = T_1 e^{\mu\beta}$ (β in rad)
Belt power	$P = (T_2 - T_1) v$
Collar bearing	$M = \frac{2}{3} \mu P (R_2^3 - R_1^3)/(R_2^2 - R_1^2)$
Journal bearing	$M \approx \mu P r$, Power $= M \omega = M \cdot 2\pi N/60$
Rolling resistance	$P \approx W a / r$

Worked example: Turns on a bollard

A ship surges against a mooring line with 20 kN. A deckhand can hold 200 N on the tail. With μ = 0.3 between the rope and the bollard, how many turns are needed?

1. T_2 = 20 000 N (load side) and T_1 = 200 N (deckhand). T_2/T_1 = 100.
2. 100 = $e^{0.3\beta}$, so β = ln(100)/0.3.
3. β = 4.605/0.3 = **15.4 rad**
4. Turns = β/2π = **2.44**, so take 3 full turns. Each extra turn multiplies the holding power by $e^{0.3 \cdot 2\pi} \approx 6.6$.

Answer: β = 15.4 rad ≈ 2.44 turns, so use 3.

ON THE SHIP

This is the most directly 'ship' week. Bollards, capstans and windlass warping ends use the capstan equation. V-belts drive engine-room auxiliaries. Screw jacks and jacking bolts are used for engine alignment. The propeller thrust block is a collar bearing, and line-shaft bearings are journal bearings. Belt power and bearing power loss are both listed Applied Mechanics outcomes.

Book problems

☐	**Screws**	8–71, 8–72, 8–77	book pp. 437–438; answers p. 658
☐	**Belts**	8–83, 8–84, 8–89, 8–91	book pp. 442–443; answers p. 658
☐	**Bearings**	8–107, 8–108, 8–116, 8–117, 8–120	book pp. 454–456; answers p. 658
☐	**Review**	R8–4, R8–6, R8–8	book p. 462; solutions p. 648

WEEK 10 · CH. 9

Centre of gravity, centroids and fluid pressure

BY THE END OF THIS WEEK YOU CAN

- Find the centroid of composite areas, including holes
- Find the centre of gravity of a set of weights, including a ship's KG
- Use Pappus to get areas and volumes of revolution
- Calculate hydrostatic pressure and the force on submerged plane surfaces
- Locate the centre of pressure

§	READING	PAGES	
9.1	Centre of gravity, centre of mass, centroid	book pp. 465–487 (PDF pp. 487–509)	SKIM
9.2	Composite bodies	book pp. 488–501 (PDF pp. 510–523)	CORE
9.3	Theorems of Pappus and Guldinus	book pp. 502–510 (PDF pp. 524–532)	SKIM
9.4–9.5	General distributed loads, fluid pressure	book pp. 511–528 (PDF pp. 533–550)	CORE

	SESSION	WHAT TO DO	TIME
☐	**Read**	Core: §9.2, 9.4–9.5. Skim: §9.1, 9.3. Copy each formula onto a cheat card.	90 min
☐	**Notes + worked example**	Work through the notes, then try the worked example before reading the steps.	90 min
☐	**Book problems**	P9–1; F9–7 to F9–12 (composite), F9–13 to F9–16 (Pappus), F9–17 to F9–21 (fluids); R9–1, R9–2, R9–5, R9–8 to R9–10. Check every answer in the back of the book.	60 min
☐	**Weekly test**	10 questions, about 40 minutes, closed book. Pass mark 70%.	60 min

Notes

Centroid and centre of gravity

The **centre of gravity** G is the point where the total weight acts. The **centroid** C is the geometric centre of a shape. The two coincide for a uniform material. You can find them by integration (section 9.1), but in practice you build shapes from standard pieces.

Composite method

Break the shape into simple parts. Tabulate each part's area A and centroid position $\tilde{x}$, $\tilde{y}$. Then $\bar{x} = \Sigma\tilde{x}A / \Sigma A$ and $\bar{y} = \Sigma\tilde{y}A / \Sigma A$. A hole is a **negative area**. Standard centroids: rectangle at the middle; triangle at h/3 from the base; semicircle at 4r/3π from the diameter; quarter circle at 4r/3π from each straight edge.

A ship's KG

The vertical centre of gravity of a loaded ship works the same way with weights: $KG = \Sigma(w \times Kg) / \Sigma w$. Kg is each item's height above the keel. Loading weight low pulls KG down. Discharging from low down pushes KG up.

Pappus–Guldinus

Spin a line or area about an axis. Surface area = θ $\bar{r}$ L and volume = θ $\bar{r}$ A, where $\bar{r}$ is the distance from the axis to the centroid of the line or area, and θ = 2π for a full turn. This is quick for rings, gaskets and V-belt pulleys.

Fluid pressure

Pressure at depth h is $p = \rho g h$. It acts perpendicular to any surface. Use ρ = 1000 kg/m³ for fresh water and 1025 kg/m³ for seawater.

Force on a submerged plane

Resultant **$F = \rho g \bar{h} A$**, where $\bar{h}$ is the depth of the plate's *centroid*. It acts at the **centre of pressure**, which is always below the centroid: $h_{cp} = \bar{h} + I_G / (\bar{h}A)$. For a vertical rectangle whose top edge is at the surface, the centre of pressure is at **⅔ of the depth**, from the triangular load in week 4.

FORMULA CARD

Composite centroid	$\bar{x} = \Sigma \tilde{x}A \;/\; \Sigma A,\ \bar{y} = \Sigma \tilde{y}A \;/\; \Sigma A$
Ship's KG	$KG = \Sigma(w \cdot Kg) \;/\; \Sigma w$
Triangle / semicircle	$h/3$ from base, $4r / 3\pi$ from diameter
Pappus	$A = \theta\, \bar{r}\, L,\ V = \theta\, \bar{r}\, A$
Hydrostatic pressure	$p = \rho\, g\, h$
Force on plane surface	$F = \rho\, g\, \bar{h}\, A$
Centre of pressure	$h_{cp} = \bar{h} + I_G \;/\; (\bar{h}\, A)$ (rectangle: $I_G = bd^3/12$)

Worked example: Load on a tank-end bulkhead

A fresh-water tank has a vertical end bulkhead 3 m wide, and the water is 2 m deep. Find the resultant force on the bulkhead and how far below the surface it acts.

1. Wetted area $A = 3 \times 2 = 6$ m². Centroid depth $\bar{h} = 1$ m.
2. $F = \rho g \bar{h} A = 1000 \times 9.81 \times 1 \times 6 =$ **58.9 kN**.
3. Centre of pressure: $I_G = 3 \times 2^3/12 = 2$ m⁴, so $h_{cp} = 1 + 2/(1 \times 6) =$ **1.333 m** below the surface.
4. Check: the pressure diagram is a triangle, and its resultant acts at ⅔ × 2 = 1.333 m ✓

Answer: F = 58.9 kN, acting 1.333 m below the surface (0.667 m above the tank bottom).

ON THE SHIP

Centre of gravity and hydrostatic force on immersed plane surfaces are both named Applied Mechanics outcomes. On board, the same maths gives the ship's KG in the stability book, loads on tank boundaries and sea-chest covers, the force on a watertight door, and the head a pump has to work against.

Book problems

☐	**Preliminary**	P9–1	book p. 479; solutions p. 638
☐	**Fundamental**	F9–7 to F9–12 (composite), F9–13 to F9–16 (Pappus), F9–17 to F9–21 (fluids)	book pp. 493, 506, 519; solutions p. 620
☐	**Review**	R9–1, R9–2, R9–5, R9–8 to R9–10	book p. 526; solutions p. 648

WEEK 11 · CH. 10 & 11

Moments of inertia and stability

BY THE END OF THIS WEEK YOU CAN

- Calculate I for rectangles and circles, and the polar moment J for shafts
- Use the parallel-axis theorem for composite sections
- Find the radius of gyration
- Calculate mass moments of inertia for discs, flywheels and rods
- Judge whether an equilibrium is stable, neutral or unstable

§	READING	PAGES	
10.1–10.3	Area moment of inertia, parallel-axis theorem, radius of gyration	book pp. 529–539 (PDF pp. 551–561)	CORE
10.4	Composite areas	book pp. 540–547 (PDF pp. 562–569)	CORE
10.8	Mass moment of inertia	book pp. 563–580 (PDF pp. 585–602)	CORE
10.5–10.7	Product of inertia, inclined axes, Mohr's circle	book pp. 548–562 (PDF pp. 570–584)	SKIP
11.1–11.3	Virtual work	book pp. 581–596 (PDF pp. 603–618)	SKIM
11.4–11.7	Potential energy and stability of equilibrium	book pp. 597–615 (PDF pp. 619–637)	CORE

	SESSION	WHAT TO DO	TIME
☐	**Read**	Core: §10.1–10.3, 10.4, 10.8, 11.4–11.7. Skim: §11.1–11.3. Copy each formula onto a cheat card.	90 min
☐	**Notes + worked example**	Work through the notes, then try the worked example before reading the steps.	90 min
☐	**Book problems**	F10–1 to F10–8; F11–1 to F11–3 (optional); R10–1 to R10–4, R10–8. Check every answer in the back of the book.	60 min
☐	**Weekly test**	10 questions, about 40 minutes, closed book. Pass mark 70%.	60 min

Notes

Why I matters

The area moment of inertia I measures how well a cross-section resists bending. Bending stress is $\sigma = My/I$ and torsional shear is $\tau = Tr/J$. Those formulas come in the 'beams' and 'torsion' parts of Applied Mechanics, which are outside this book. You need I and J ready for them.

Standard results

Rectangle, width b and depth h, about its own centroid: $\mathbf{\bar{I} = bh^3/12}$. About its base: $bh^3/3$. Solid circle: $I = \pi d^4/64$, and the polar moment is $\mathbf{J = \pi d^4/32}$. Hollow shaft: $J = \pi(D^4 - d^4)/32$. Notice the depth is **cubed**: doubling the depth makes a beam 8 times stiffer.

Parallel-axis theorem

$\mathbf{I = \bar{I} + A\,d^2}$, where d is the distance from the shape's own centroid to the new axis. The centroidal value is always the **smallest**. For a composite section: (1) find the overall centroid (week 10), (2) add $\bar{I} + Ad^2$ for each part, (3) subtract the holes.

Radius of gyration

$k = \sqrt{I/A}$. This is the distance at which all the area could be concentrated and still give the same I. Column (strut) formulas use it. For mass, $k = \sqrt{I/m}$.

Mass moment of inertia

$I = \int r^2 \, dm$ measures resistance to *angular acceleration*. It's the rotational version of mass. Solid disc or flywheel: **$\frac{1}{2}mr^2$**. Thin ring: mr^2. Slender rod about its centre: $mL^2/12$, and about one end: $mL^2/3$. Parallel axis: $I = I_G + md^2$. You'll need these for flywheels, propellers and rotors in dynamics.

Stability of equilibrium

With potential energy V (gravity plus springs) as a function of position θ, equilibrium is where **$dV/d\theta = 0$**. If $d^2V/d\theta^2 > 0$ (V at a minimum), the equilibrium is **stable**: nudge it and it returns. If $d^2V/d\theta^2 < 0$ it is **unstable**, and if V is constant it is **neutral**. A pendulum hanging down is stable. A ship is stable when its centre of gravity G is below its metacentre M, so the metacentric height GM is positive. Raising KG (week 10) reduces GM.

FORMULA CARD

Rectangle	$\bar{I} = bh^3/12$, $I_{base} = bh^3/3$
Circle / polar	$I = \pi d^4/64$, $J = \pi d^4/32$
Hollow shaft	$J = \pi(D^4 - d^4)/32$
Parallel axis	$I = \bar{I} + A\,d^2$ (mass: $I = I_G + m\,d^2$)
Radius of gyration	$k = \sqrt{I/A}$ or $\sqrt{I/m}$
Disc / ring / rod	$\frac{1}{2}mr^2$, mr^2, $mL^2/12$ (centre), $mL^2/3$ (end)
Stability	$dV/d\theta = 0$; $d^2V/d\theta^2 > 0$ stable

Worked example: Bending stiffness of a T-bar stiffener

Use the T-section from week 10 (web 20 × 180 mm, flange 200 × 20 mm on top). Find I about the horizontal axis through its centroid.

1. From week 10: $\bar{y} = 142.6$ mm above the bottom.
2. Web: $\bar{I} = 20 \times 180^3/12 = 9.72 \times 10^6$ mm⁴. $d = 142.6 - 90 = 52.63$ mm. $Ad^2 = 3600 \times 52.63^2 = 9.97 \times 10^6$ mm⁴.
3. Flange: $\bar{I} = 200 \times 20^3/12 = 0.133 \times 10^6$ mm⁴. $d = 190 - 142.6 = 47.37$ mm. $Ad^2 = 4000 \times 47.37^2 = 8.98 \times 10^6$ mm⁴.
4. Total: $I = 9.72 + 9.97 + 0.133 + 8.98 =$ **28.8×10^6 mm⁴**.

Answer: $I = 28.8 \times 10^6$ mm⁴ about the centroidal axis.

ON THE SHIP

I and J feed straight into beam bending and shaft torsion: line-shaft sizing, coupling bolts and stiffener strength are all on the Applied Mechanics syllabus. Mass moment of inertia governs flywheels and turning gear. The stability section is the statics behind GM and the ship's stability book.

Book problems

☐	**Fundamental**	F10–1 to F10–8	book pp. 536 & 543; solutions p. 620
☐	**Fundamental**	F11–1 to F11–3 (optional)	book p. 591; solutions p. 620
☐	**Review**	R10–1 to R10–4, R10–8	book p. 578; solutions p. 648

WEEK 12 · ALL CHAPTERS

Review and final exam

BY THE END OF THIS WEEK YOU CAN

- Pull the whole course together under exam conditions
- Find your weak topics from the weekly test scores and fix them
- Sit a 20-question final in 90 minutes and aim for 70% or more

§	READING	PAGES	
Review	Re-read your formula sheet and every 'Important Points' box	All chapters	CORE
Ch. 2–10	Chapter Review summaries at the end of each chapter	End of each chapter	CORE

	SESSION	WHAT TO DO	TIME
☐	**Fix weak spots**	Redo every weekly-test question you missed, without looking at the explanation first.	90 min
☐	**Formula sheet**	Rewrite the full formula sheet from memory, then check it.	90 min
☐	**Timed review problems**	R5–1, R5–2, R7–3, R7–4, R8–3, R8–6, R9–8, R9–9	60 min
☐	**Final exam**	20 questions in one sitting, closed book. Aim for 70% or more.	90 min

Notes

How to use this week

Sessions 1–2: go through the hub's progress page and redo every weekly-test question you got wrong, without looking at the explanation first. Session 3: work two Review problems from each of chapters 5, 7, 8 and 9 under time pressure. Session 4: sit the final exam in one go, closed book, with a calculator and your formula sheet only.

Exam technique

Draw a free-body diagram for *every* question, even the 'easy' ones. Write the equation before the numbers. Keep units on every line. Round only at the end, to 3 significant figures. If an answer looks odd (a 5 kg pump needing 9 kN to move), go back to the FBD first.

What this book doesn't cover

The Applied Mechanics exams also include **dynamics** (linear and angular motion, work, energy and power, which is Hibbeler's Dynamics, chapters 12–22), **strength of materials** (stress and strain, beams, torsion, thin cylinders), **lifting machines** (mechanical advantage, velocity ratio, efficiency) and **hydraulics** (flow through pipes and orifices). Finishing this plan gives you the statics foundation every one of those topics stands on. A follow-on plan for dynamics is the natural next step.

> ON THE SHIP
>
> The final mixes every week's topics in the style of the Applied Mechanics papers. You'll see concurrent forces, beams, friction on planes and in bearings, belts, centre of gravity and hydrostatic force. Treat it as a mock exam.

Book problems

☐	**Review (timed)**	R5–1, R5–2, R7–3, R7–4, R8–3, R8–6, R9–8, R9–9	solutions p. 648

ALL WEEKS

Formula sheet

WEEK 1 · UNITS, VECTORS AND ADDING FORCES

Weight	$W = m\,g,\ g = 9.81\ \text{m/s}^2$
Cosine rule	$C = \sqrt{(A^2 + B^2 - 2AB\cos c)}$
Parallelogram (angle θ between the forces)	$R = \sqrt{(A^2 + B^2 + 2AB\cos\theta)}$
Sine rule	$A / \sin a = B / \sin b = C / \sin c$
Components	$F_x = F\cos\theta,\ F_y = F\sin\theta$
Resultant	$F_R = \sqrt{((\Sigma F_x)^2 + (\Sigma F_y)^2)},\ \theta = \tan^{-1}(\Sigma F_y / \Sigma F_x)$
Equilibrant	$E = F_R$, direction $\theta_R + 180°$

WEEK 2 · EQUILIBRIUM OF A PARTICLE AND FREE-BODY DIAGRAMS

Particle equilibrium	$\Sigma F_x = 0,\ \Sigma F_y = 0$ ($\Sigma F_z = 0$ in 3D)
Lami's theorem	$F_1/\sin\alpha_1 = F_2/\sin\alpha_2 = F_3/\sin\alpha_3$
Symmetric two-leg sling	$T = W / (2\cos\theta)$ (θ from vertical)
Linear spring	$F = k\,s,\ s = L - L_0$
Force along a line	$F = F\cdot(r / \lvert r\rvert)$
Direction cosines	$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$
Dot product	$A\cdot B = AB\cos\theta = A_xB_x + A_yB_y + A_zB_z$

WEEK 3 · MOMENTS OF A FORCE

Scalar moment	$M_O = F\,d$ (d ⊥ to line of action)
Force at an angle φ to a lever of length L	$M = F\,L\sin\varphi$
Varignon (2D)	$M_O = x\,F_y - y\,F_x$
Resultant moment	$(M_R)_O = \Sigma\,F\,d$ (CCW +)
Vector moment	$M_O = r \times F$
Moment about an axis	$M_a = u_a \cdot (r \times F)$

WEEK 4 · COUPLES, EQUIVALENT SYSTEMS AND DISTRIBUTED LOADS

Couple moment	$M = F\,d$ (free vector)
Force moved to O	F at O + couple $M = F\,d$
Parallel resultant	$F_R = \Sigma F,\ \bar{x} = \Sigma(F\,x) / \Sigma F$
Uniform load	$F = wL$ at L/2
Triangular load	$F = \tfrac{1}{2}\,w_0\,L$ at L/3 from the w_0 end
Any distributed load	$F_R = \int w\,dx,\ \bar{x} = \int x\,w\,dx / \int w\,dx$

WEEK 5 · EQUILIBRIUM OF A RIGID BODY

2D equilibrium	$\Sigma F_x = 0,\ \Sigma F_y = 0,\ \Sigma M_O = 0$
Roller / pin / fixed	1 / 2 / 3 unknowns (fixed adds M)
Simply supported, point load P at a from A (span L)	$R_B = P\,a / L,\ R_A = P\,(L - a)/L$
Cantilever, end load P	$R = P,\ M_{wall} = P\,L$
Cantilever, UDL w	$R = wL,\ M_{wall} = wL^2/2$
Lever	$F_1\,d_1 = F_2\,d_2$

WEEK 6 · TRUSSES, FRAMES AND MACHINES

Each joint	$\Sigma F_x = 0,\ \Sigma F_y = 0$
Each section	$\Sigma F_x = 0,\ \Sigma F_y = 0,\ \Sigma M = 0$ ($\leq$ 3 cut members)
Member force from its y-part	$F = F_y \times$ (length / rise)
Chord force by sections	$F_{chord} = M_{section}$ / depth
Mechanical advantage	$MA = F_{out} / F_{in}$

WEEK 7 · INTERNAL FORCES: SHEAR FORCE AND BENDING MOMENT DIAGRAMS

Load–shear	$dV/dx = w$ (jump of P at a point load)
Shear–moment	$dM/dx = V$, ΔM = area under V diagram
SS beam, central P	$M_{max} = PL/4$
SS beam, UDL	$M_{max} = wL^2/8,\ V_{max} = wL/2$
SS beam, P at a from A	$M_{max} = P\,a\,(L - a) / L$
Cantilever, end P	$M_{max} = PL,\ V = P$
Cantilever, UDL	$M_{max} = wL^2/2,\ V_{max} = wL$

WEEK 8 · DRY FRICTION AND WEDGES

Static (limit) / kinetic	$F_s = \mu_s N,\ F_k = \mu_k N$
Angle of friction	$\tan\varphi = \mu$
Push up a slope (P ∥ slope)	$P = W\,(\sin\theta + \mu\cos\theta)$
Hold on a slope (P ∥ slope)	$P = W\,(\sin\theta - \mu\cos\theta)$
Work up a slope	$U = P\,s$
Tipping (push at height h, base b)	$P_{tip} = W\,b / (2h)$

WEEK 9 · SCREWS, BELTS, BEARINGS AND ROLLING RESISTANCE

Screw lead angle	$\theta = \tan^{-1}(l / 2\pi r)$
Screw raise / lower	$M = W r \tan(\theta + \varphi)$, $M = W r \tan(\varphi - \theta)$
Self-locking	$\varphi_s > \theta$
Belt / capstan	$T_2 = T_1 e^{\mu\beta}$ (β in rad)
Belt power	$P = (T_2 - T_1) v$
Collar bearing	$M = \frac{2}{3} \mu P (R_2^3 - R_1^3)/(R_2^2 - R_1^2)$
Journal bearing	$M \approx \mu P r$, $Power = M \omega = M \cdot 2\pi N/60$
Rolling resistance	$P \approx W a / r$

WEEK 10 · CENTRE OF GRAVITY, CENTROIDS AND FLUID PRESSURE

Composite centroid	$\bar{x} = \Sigma \tilde{x} A / \Sigma A$, $\bar{y} = \Sigma \tilde{y} A / \Sigma A$
Ship's KG	$KG = \Sigma(w \cdot Kg) / \Sigma w$
Triangle / semicircle	$h/3$ from base, $4r / 3\pi$ from diameter
Pappus	$A = \theta \bar{r} L$, $V = \theta \bar{r} A$
Hydrostatic pressure	$p = \rho g h$
Force on plane surface	$F = \rho g \bar{h} A$
Centre of pressure	$h_{cp} = \bar{h} + I_G / (\bar{h} A)$ (rectangle: $I_G = bd^3/12$)

WEEK 11 · MOMENTS OF INERTIA AND STABILITY

Rectangle	$\bar{I} = bh^3/12$, $I_{base} = bh^3/3$
Circle / polar	$I = \pi d^4/64$, $J = \pi d^4/32$
Hollow shaft	$J = \pi(D^4 - d^4)/32$
Parallel axis	$I = \bar{I} + A d^2$ (mass: $I = I_G + m d^2$)
Radius of gyration	$k = \sqrt{(I/A)}$ or $\sqrt{(I/m)}$
Disc / ring / rod	$\frac{1}{2} m r^2$, mr^2, $mL^2/12$ (centre), $mL^2/3$ (end)
Stability	$dV/d\theta = 0$; $d^2V/d\theta^2 > 0$ stable